

>> critical value.

$$t_{\alpha, n-1} = t_{0.05, 3} = 2.353$$

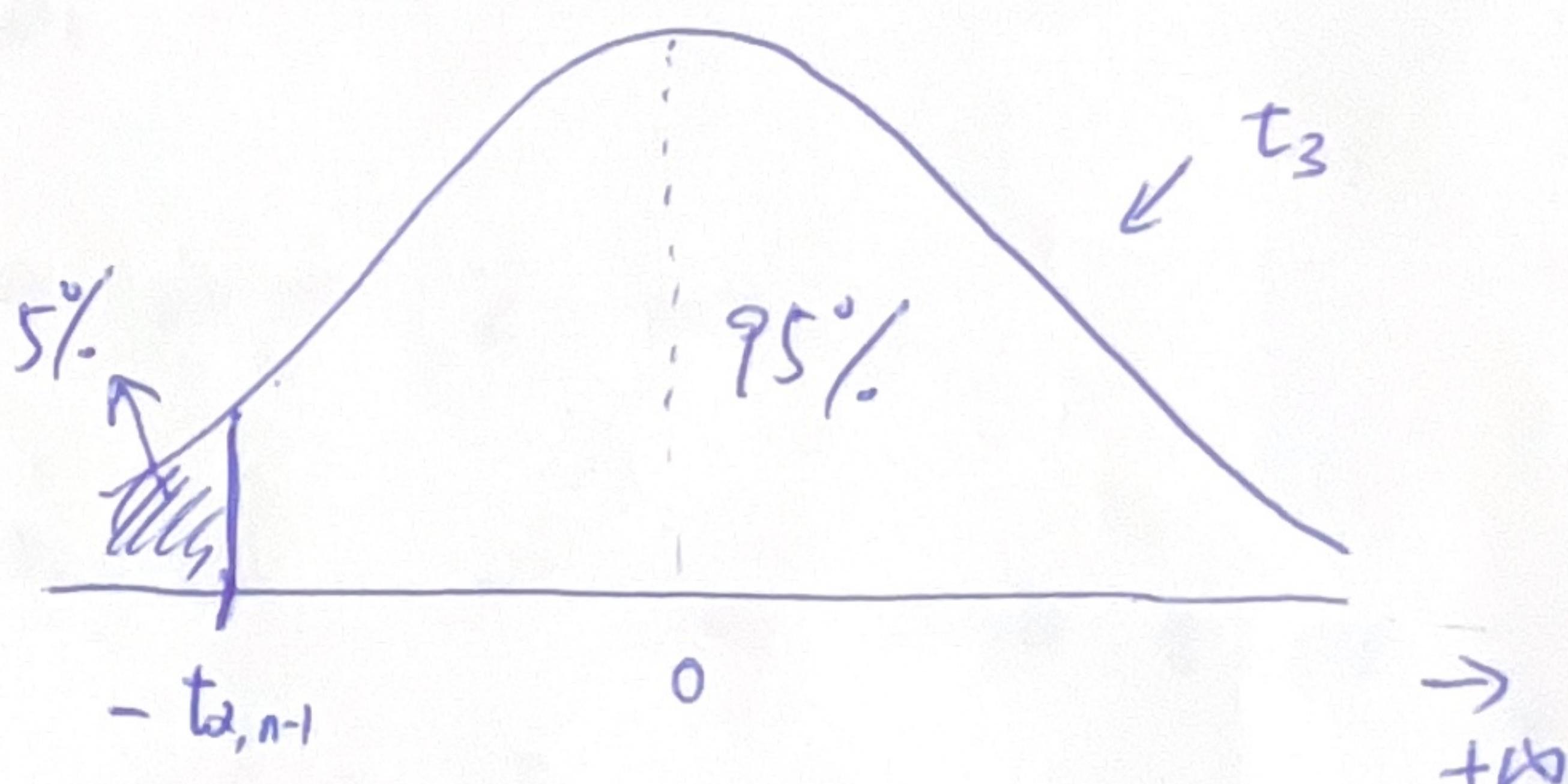
$$3> t^* = 3.286 > 2.353 = t_{\alpha, n-1}$$

Therefore, reject H_0 .

We have strong enough evidence to support $\mu > 5.3$.

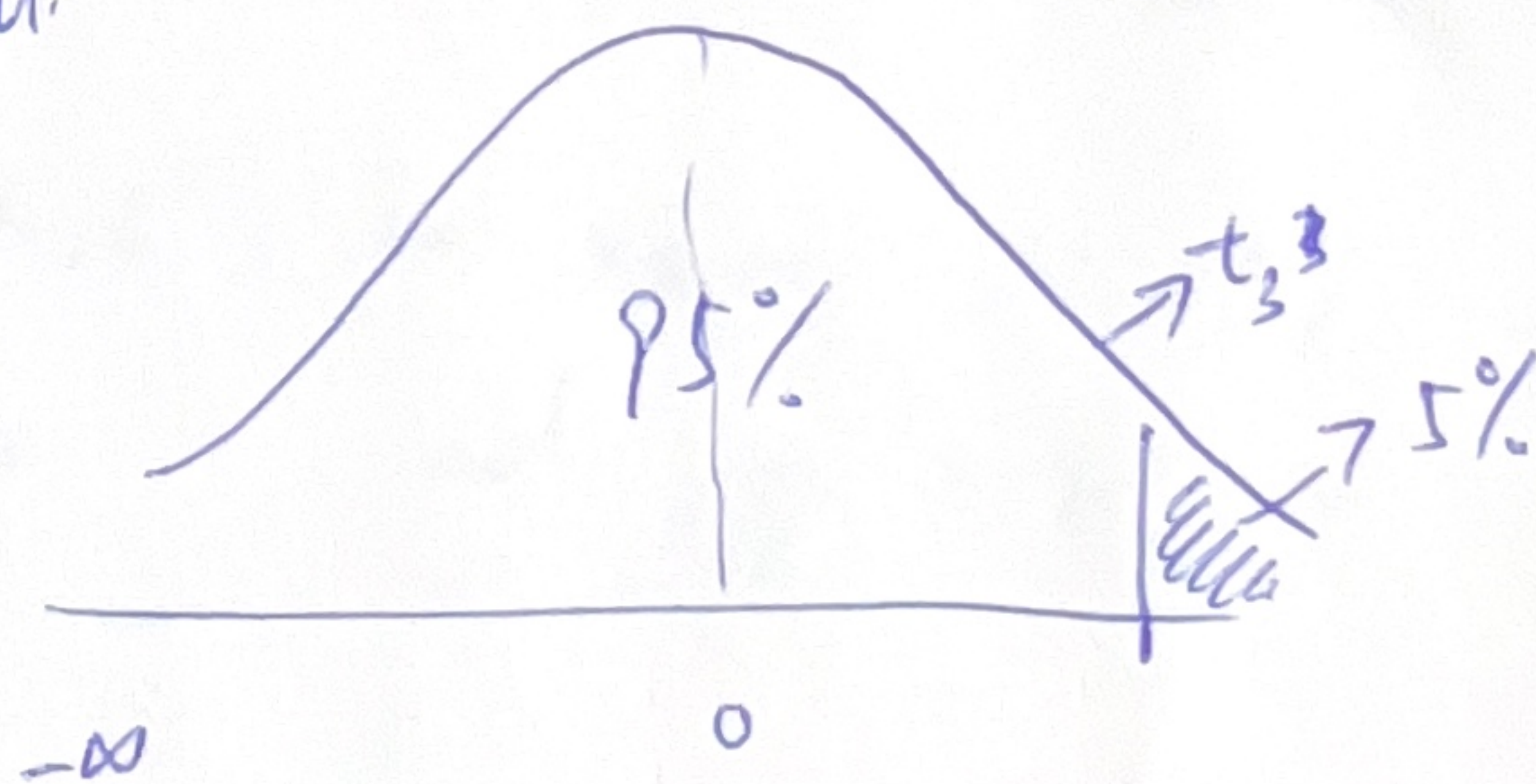
2> 95% lower bound. (1-sided CI) for μ .

$$\begin{aligned} & \left(\bar{x} - t_{\alpha, n-1} \frac{s}{\sqrt{n}}, +\infty \right) \\ &= \left(5.6 - 2.353 \times \frac{0.1826}{\sqrt{4}}, +\infty \right) \\ &= (5.385, \infty) \end{aligned}$$



1> 95% upper bound CI for μ .

$$(-\infty, \bar{x} + t_{\alpha, n-1} \frac{s}{\sqrt{n}})$$



Review: one sample test for μ .

1. σ - known $\Rightarrow$ one sample Z-test.

2. σ - unknown $\Rightarrow$ one sample t-test.

Both of them require $\left\{ \begin{array}{l} \textcircled{1} X \sim N(\mu, \sigma^2) \\ \text{or} \\ \textcircled{2} X \sim \cdot(\mu, \sigma^2), n \geq 30. \end{array} \right.$

Two sample test for $\mu_1 - \mu_2$.

1. σ_1, σ_2 known $\Rightarrow$ ~~one~~ two sample Z-test.

2. σ_1, σ_2 unknown, assume $\sigma_1 = \sigma_2 = \sigma$. $\Rightarrow$ two sample t-test.

Both of them require $\left\{ \begin{array}{l} \textcircled{1} X_1 \sim N(\mu_1, \sigma_1^2) \text{ and } X_2 \sim N(\mu_2, \sigma_2^2) \\ \text{or} \\ \textcircled{2} X_1 \sim \cdot(\mu_1, \sigma_1^2) \text{ and } X_2 \sim \cdot(\mu_2, \sigma_2^2) \\ n_1 \geq 30 \quad \text{and} \quad n_2 \geq 30 \end{array} \right.$

Hypothesis test for $\mu_1 - \mu_2$ with σ_1, σ_2 known.

(4)

$$H_0: \mu_1 - \mu_2 = d_0$$

$\downarrow$
value.

$$\text{vs. } \begin{cases} H_a: \mu_1 - \mu_2 \neq d_0 & \text{two-sided} \\ H_a: \mu_1 - \mu_2 > d_0 & \text{right-sided} \\ H_a: \mu_1 - \mu_2 < d_0 & \text{left-sided} \end{cases}$$

3-step test

1) test statistic

$$Z^* = \frac{(\bar{X}_1 - \bar{X}_2) - d_0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

2) critical value.

$$\text{two-sided: } Z_{\frac{\alpha}{2}}$$

$$\text{right-sided: } Z_{\alpha}$$

$$\text{left-sided: } -Z_{\alpha}$$

3) Decision and conclusion.

two-sided: If $|Z^*| > Z_{\frac{\alpha}{2}}$, then reject H_0

right-sided: If $Z^* > Z_{\alpha}$, then reject H_0

left-sided: If $Z^* < -Z_{\alpha}$, then reject H_0

4) P-value.

$$\text{two-sided: } P\text{-value} = 2P(Z > |Z^*|)$$

$$\text{right-sided: } P\text{-value} = P(Z > Z^*)$$

$$\text{left-sided: } P\text{-value} = P(Z < Z^*)$$

if p-value $< \alpha$, then reject H_0

57. $(1 - \alpha) 100\%$ CI for $(\mu_1 - \mu_2)$

$$(\bar{x}_1 - \bar{x}_2) \pm Z_{\frac{\alpha}{2}} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$